BT501 Paper Solved
Bt501 mcqs
allele-specific PCR. The amplification of **specific alleles**, or DNA sequence variants, at the same locus by polymerase chain reaction.
Real-Time PCR a specialized technique that allows a PCR reaction to be visualized "in real time" as the reaction progresses.
Real-time PCR for infectious disease
TaqMan probes are hydrolysis probes that are designed to increase the specificity of quantitative **PCR**

- High sensitivity: More than one labeled antibody is bound per antigen molecule;
- Flexible: Different primary detection antibodies can be used with a single labeled secondary antibody;
- Cost-saving: Fewer labeled antibodies are required.

- Enzyme Linked Immunosorbent Assay (ELISA)
- Term Was Coined By Engvall and Pearlmann in 1971

DNA-based detection methods for detection of penaeid shrimp viruses are now used routinely in a number of laboratories around the world

Genetic disorders may also be complex, multifactorial, or **polygenic**, meaning they are likely associated with the effects **of** multiple **genes** in combination with lifestyles and environmental factors.

- **Computational** methods typically use computer programs to analyze proteins. However, many experimental methods (e.g., mass spectrometry) require computational analysis of the raw data.

Qualitative **immunoassays** are often used to detect antigens on **infectious agents**and antibodies that the body produces to fight them.

oligonucleotide ligation assay (OLA)

- Mitochondrial homeostasis is regulated by a balance between mitochondrial biogenesis and degradation.

Mitochondrial disorders
Cardiovascular disease generally refers to conditions that involve narrowed or blocked blood vessels that can lead to a heart attack, chest pain (angina) or stroke. Other heart conditions, such as those that affect your heart's muscle, valves or rhythm, also are considered forms of heart disease

- In 2005 CVD accounted for approximately 38 percent of all deaths

- **Padlock probes** (PLPs) are long oligonucleotides, whose ends are complementary to adjacent target sequences. Upon hybridization to the target, the two ends are brought into contact, allowing PLP circularization by ligation

- Parkinson’s disease is a disorder that affects nerve cells in the part of the brain controlling muscle movement

Papers Answers
What is diabetes mellitus
Diabetes mellitus (DM) is a group of diseases characterized by high levels of blood glucose resulting from defects in insulin production, insulin action, or both.
The term diabetes mellitus describes a metabolic disorder of multiple aetiology characterized by chronic hyperglycaemia with disturbances of carbohydrate, fat and protein metabolism resulting from defects in insulin secretion, insulin action, or both.
The effects of diabetes mellitus include long–term damage, dysfunction and failure of various organs.

How to diagnose polygenic disease
• Examples of polygenic conditions include hypertension, coronary heart disease, and diabetes. Because such disorders depend on the simultaneous presence of several genes, they are not inherited as simply as are single- gene diseases.
Many genetic markers that relate to common multifactorial disease in adults have been identified during the past 15 years. Their use as adjuncts for the diagnosis, prognosis, prediction of disease or targeting therapy for these disorders has begun, good examples being the Factor V Leiden mutation for venous- thromboembolism, lipoprotein lipase mutations for hypertriglyceridaemia and the apolipoprotein E4 variant for Alzheimer's dementia. However, extensive gene–gene and gene–environment interactions make their use more complex than markers for the simpler monogenic disorders
Despite the uncertainties associated with environmental factors, GWAS data and gene-disease networking represent significant steps in the journey toward a fuller understanding of the role of genetics in human disease.
As a result of these efforts, associations between seemingly unrelated diseases are coming to light, and researchers are discovering treatment options that might not have otherwise been considered.
Although a shift in the traditional diagnosis and treatment of disease by health care professionals is still a long way off, information regarding molecular pathways and genetic correlations is already proving successful in the treatment of some conditions, and patients stand to benefit even further from

continued research in the field.
• Genetic disorders may also be complex, multifactorial, or polygenic, meaning they are likely associated with the
effects of multiple genes in combination with lifestyles and environmental factors. Multifactorial disorders include
heart disease and diabetes.

What is mitochondrial disease?2
If a child is stricken with a catastrophic disease affecting three or more organ systems, or if a child has been afflicted with a relapsing disease that affects two or more organ systems and leads to slow but measurable deterioration, he or she may have a mitochondrial disease.

Type 2 diabetes?3
Type 2 is the most common form of diabetes (DM).
In this form, the body does not produce enough insulin or the cells ignore the insulin that is produced.
Insulin is required by the body in order to use sugar.
Insulin takes the sugar from the blood into the cells.
When glucose builds up in the blood instead of going into cells, it can cause some problems:
– In the short-term, your cells may be starved for energy
– In time, high blood glucose may hurt your eyes, kidneys, nerves, or heart.

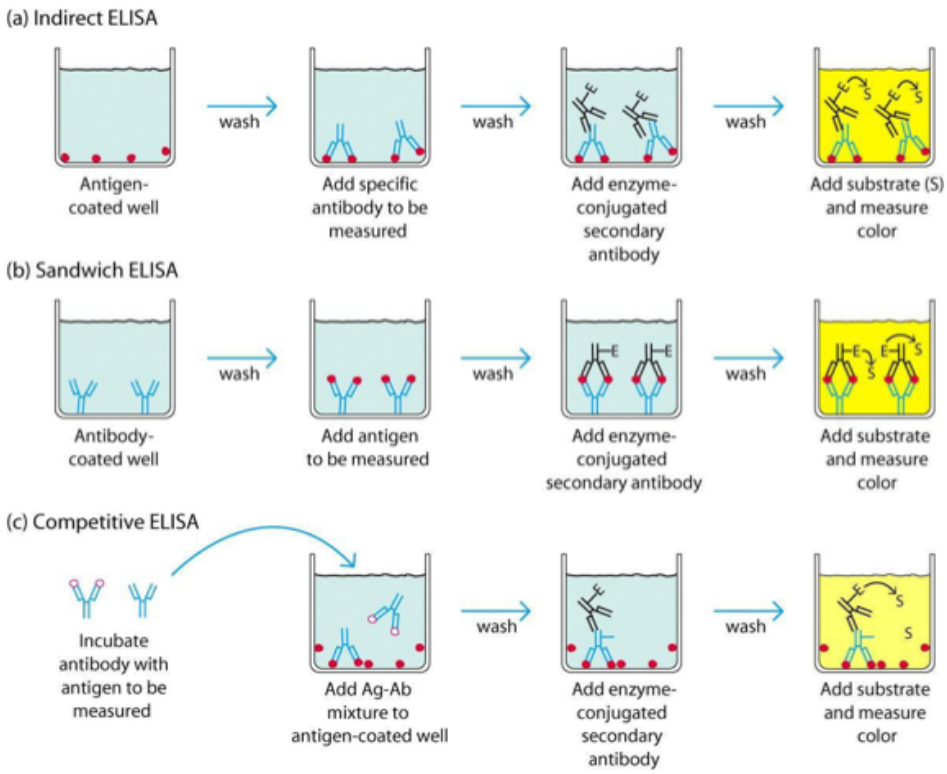
What is taqman pcr?3
TaqMan probes are hydrolysis probes that are designed to increase the specificity of quantitative PCR. ... TaqMan probes were named after the videogame Pac-Man (Taq Polymerase + PacMan = TaqMan) as it's mechanism is based on the Pac-Man principle.
taqMan probes consist of a
fluorophore covalently attahed to the 5′-end of the oligonucleotide probe and a quencher at the 3′- end
TaqMan probes are designed such that they anneal within a DNA region amplified by a specific set of primers. (Unlike the diagram, the probe binds to single stranded DNA.) As the Taq polymerase extends
the primer and synthesizes the nascent strand (again, on a single-strand template, but in the direction opposite to that

Immunoassay for infectious disease?3
Qualitative immunoassays are often used to detect antigens on infectious agentsand antibodies that the body produces to fight them. For example, immunoassaysare used to detect antigens on Hemophilus, Cryptococcus , and Streptococcus organisms in the cerebrospinal fluid (CSF) of meningitis patients.

What is real time PCR?5

The Polymerase Chain Reaction (PCR) is a process for the amplification of specific fragments of DNA.
Real-Time PCR a specialized technique that allows a PCR reaction to be visualized “in real time” as the reaction progresses.
As we will see, Real-Time PCR allows us to measure minute amounts of DNA sequences in a sample!

Describe the procedure of Eliza?5
Enzyme Linked Immunosorbent Assay (ELISA)
Term Was Coined By Engvall and Pearlmann in 1971
DifferentTypes
– Sandwich
– Indirect
– Competitive
Similar To RIA, Except No Radiolabel
Can Be Used To Detect Both Antibody and Antigen
VerySensitive,pg/mL
Relies on Monoclonal Abs



Describe the method of OLA?5

Oligonucleotide Ligation Assay (OLA)

- A pair of oligonucleotides designed to anneal to adjacent sequences within a PCR product → If perfectly hybridized → joined by a DNA ligase.
- Oligonucleotides complementary to normal and mutant sequences are differentially labeled and the products are identified by a computer software.

A method to identify a specific oligonucleotide sequence without use of radiochemicals, electrophoresis or centrifugation. An oligonucleotide biotinylated at its 5'-end and another with a reporter group (chromophore or fluorophore) at its 3'- end are constructed to hybridize to the sequence to be detected in a template DNA strand.
A ligase is able to join the two oligonucleotides if they both match the template perfectly. The ligated oligonucleotide can be extracted from the mixture on to immobilized streptavidin and the presence of the reporter group detected.

Parkinson's disease?10
Background
Parkinson’s disease is a disorder that affets nerve cells in the part of the brain controlling muscle movement
Disease is progressive – signs/symptoms worsen over time
Eventually is disabling, but progresses gradually
Believed to be caused by genetics, environmental factors or a combination of the two
Parkinson’s Disease Stats
• First desribed by James Parkinson in 1817
• Affects ~1 million in the U.S.
• Onset typically between 50-60 years of age, and slowly progresses with age
• Average onset is 62.4 years of age
Neurological Basis
• Neurodegenerative Disease : aused y degeneration (dysfunction and death)

of neurons within the brain (nigrastriatal pathway of the basal ganglia)
• NORMALBRAINFUNCTION–BasalGanglia • Cells in substantia nigra produce/ release dopamine
• Dopamine released by SN neurons lands on neurons of other brain centers, controlling their firing
• Main targets are caudate nucleus and putamen (striatum)
• This basal ganglia pathway is involved in regulation of movement

Treatment of Parkinson's Disease
• Medicines - Can control symptoms effectively
for 5 to10 years in most people before disabling side-effects appear and control then becomes less effective
• Surgery - Deep Brain Stimulation (DBS) is an alternative to earlier forms of surgery where
lesions are cut in the area of the brain that controls fine movement of the limbs to stop shaking
• Even if surgery is successful, it only relieves symptoms for a period of time; it does not halt progression of the disease